You might have already heard about Apple switching from Intel processors to ARM processors. The reason for this is that custom-built ARM processors achieve higher performance with lower energy consumption. This is, of course, exciting not only for laptops but also for servers in the data center.

AWS offers machines based on custom-built ARM processors called *Graviton* as well. As a customer, you will notice similar performance at lower costs. However, you need to make sure that the software you want to run is compiled for the ARM64 architecture. We migrated workloads from EC2 instances with Intel processors to virtual machines with ARM processors a few times already, typically within one to four hours. We would have liked to use Graviton instances for the third edition of this book, but, unfortunately, these were not yet part of the Free Tier at that time. We highly recommend you check out the following Graviton instance types offered by AWS:

- `T4g`—Burstable and cost-effective instance types
- `M6g`, `M6gd`—General-purpose instance types
- `C6g`, `C6gd`, `C6gn`, `C7g`—Compute-optimized instance types
- `R6g`, `R6gd`, `X2gd`—Memory-optimized instance types
- `Im4gn`, `Is4gen`—Instance types providing built-in storage
- `G5g`—A special instance type optimized for Android game streaming

Configuring the key pair for login

As an administrator of a Linux machine, you used a username and password or username and a public/private key pair to authenticate yourself in the past. By default, AWS uses a username and a key pair for authentication. That’s why the next section of the wizard asks you about defining a key pair for the EC2 instance you are going to launch. We try to avoid this approach, because it works only for a single user, and it is not